

# Infrastructure-Aware Wildfire Coordination: A Physics-Grounded Benchmark and a Value-Aware Hierarchical Multi-Agent Reinforcement Learning Method

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## Abstract

Coordinating multi-agent teams to defend critical infrastructure during wildfires couples strategic asset prioritization with local suppression. We contribute (i) an *infrastructure-aware wildfire-coordination benchmark* built on the validated Cell2Fire physics engine and two documented, GIS-derived landscapes—Saudi Arabian petroleum infrastructure and California Wildland–Urban Interface (WUI) zones—under a suppression-relevant regime with a uniform 5-seed  $\times$  15-episode protocol; and (ii) **HierComm**, a *value-aware hierarchical* multi-agent policy that composes a rule-based value-aware asset assignment with a learned tactical policy trained by imitation of a value-aware expert and reinforcement-learning fine-tuning. Because the strongest baseline differs by region, consistent performance across both landscapes is non-trivial. HierComm attains the best average Weighted Economic Loss and Infrastructure Survival Rate in both regions; these improvements are statistically significant against the communicating baseline and the heuristic teacher and are largest on the scattered-asset (California) landscape, while on the concentrated-asset (Saudi) landscape the margin over the strongest flat baseline is within seed variance. A component ablation attributes the gains to the value-aware hierarchy, the learned tactical policy, and RL fine-tuning, and reports that explicit inter-agent communication is not a statistically significant driver at the three-agent scale studied. HierComm is also robust—strongest across an easy/medium/hard difficulty sweep and retaining a protective margin over inaction under held-out ignitions, rotated wind and asset layouts, and, with degradation, cross-region transfer. We make no claim of novelty for the underlying reinforcement-learning or communication primitives; our contributions are the benchmark, a simple and strong integrated system, an honest component analysis, and a fully reproducible release (code, checkpoints, GIS pipeline, hashed manifest).

## 1 Introduction

Wildfire suppression demands coordinated allocation of limited suppression resources across a dynamically evolving hazard landscape. When critical infrastructure—petroleum refineries, pipeline junctions, residential structures—lies in

the path of an advancing fire front, the coordination challenge intensifies: agents must *strategically prioritize* which assets to defend rather than simply minimizing total burned area. This infrastructure-defense framing distinguishes wildfire suppression from generic fire-containment problems and motivates the present work.

**Coordination under partial observability.** This prioritization is compounded by partial observability: agents with local, egocentric views must decide *which* distant high-value assets to protect, not merely suppress the nearest flames. Such prioritization can be difficult for flat decentralized policies such as MAPPO (Yu et al. 2022) and QMIX (Rashid et al. 2018), which tend to react to locally visible fire, and the benefit of explicit inter-agent communication varies with how assets are distributed in space. Hierarchical decision making offers a natural decomposition of the problem across timescales: a strategic layer assigns agents to high-value threatened assets on a slow timescale, while a tactical layer handles local suppression on a fast one. We build our method, **HierComm**, on this decomposition—a *value-aware commander* that dispatches agents toward the assets most worth defending, and a learned tactical policy, refined by reinforcement learning, that carries out suppression.

**A benchmark and a method, together.** Meaningful progress on this problem requires a testbed rigorous enough to trust and a method strong enough to matter; we contribute both. Our *infrastructure-aware coordination benchmark* combines physically grounded fire spread from the validated Cell2Fire engine, two documented GIS-derived landscapes with contrasting asset geometries (concentrated Saudi petroleum infrastructure and scattered California WUI zones), and a *suppression-relevant* regime whose difficulty was calibrated using only a no-operation baseline and a fixed heuristic—independently of the proposed method—so that skilled coordination, rather than inaction or luck, determines outcomes. Because the relative ranking of baselines shifts across these two landscapes, the benchmark rewards methods that perform consistently across differing asset geometries. Under a uniform 5-seed protocol, HierComm achieves the best average infrastructure-protection outcomes in both regions (Section 6); a component ablation attributes these gains to the value-aware hierarchy, the learned tactical policy, and reinforcement-learning fine-tuning, and finds that the

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explicit communication in the architecture does not significantly change performance at the three-agent scale we study. We further stress-test the method across difficulty regimes, held-out conditions, and cross-region transfer, where it remains protective though with degradation across regions. We do not claim novelty for the underlying reinforcement-learning or communication mechanisms; rather, benchmark and method are complementary contributions—the benchmark makes coordination skill measurable, and HierComm is a simple, reproducible system that meets that bar while its ablation clarifies *why* it works.

**Contributions.** Our contributions are:

1. An **infrastructure-aware wildfire-coordination benchmark** on the validated Cell2Fire physics engine and two documented, GIS-derived landscapes (Saudi petroleum and California WUI), under a *suppression-relevant* regime in which skilled coordination measurably protects infrastructure, with a released data pipeline and hashed manifests.
2. **HierComm**, a **value-aware hierarchical MARL** system that *composes* standard components—a rule-based value-aware asset assignment, a communicating tactical actor, behavior cloning from a value-aware expert, and PPO fine-tuning—rather than introducing a new learning algorithm. It attains the best average WEL and ISR in *both* regions, with WEL improvements over the communicating baseline (CommNet) and the heuristic teacher that are statistically significant; its margin over the strongest flat baseline is significant on California but within seed variance on Saudi.
3. A **component ablation** that attributes the gains: the hierarchy, the learned tactical policy, and RL fine-tuning each contribute significantly, whereas explicit inter-agent communication is *not* a statistically significant driver at three agents—an honest account of *where* the improvement originates rather than an assumed one.
4. A **robustness and transfer analysis, with a fully reproducible evaluation protocol**: HierComm stays best across an easy/medium/hard difficulty sweep and keeps a positive protective margin over no-operation under held-out ignitions, rotated wind, rotated asset layouts, and cross-region transfer, characterized with a falsifiable within-condition  $\Delta$ WEL metric. All results follow a uniform protocol—5 training seeds, bootstrap 95% CIs, Welch’s *t*-tests with effect sizes—with released code, checkpoints, GIS pipeline, and a hashed reproducibility manifest.

## 2 Related Work

**Wildfire Simulation and GIS-Based Planning.** Wildfire spread modeling has progressed from simple cellular automata to physics-based simulators. Cell2Fire (Pais et al. 2021) implements the Canadian Forest Fire Behavior Prediction (FBP) system with heterogeneous fuel types, wind effects, and topographic slope corrections. Operational research approaches optimize firebreak placement using mixed-integer programming (Hu and Ntaimo 2009) but

assume global state observability and static fire behavior, neither of which holds during active suppression. Our work uses Cell2Fire as a validated physics backend while framing suppression as a sequential decision problem under partial observability.

**Reinforcement Learning for Wildfire and Disaster Response.** Recent RL approaches to wildfire include UAV-based detection (Ghali, Akhloufi, and Wael Saber 2022; Partheepan, Sanati, and Hassan 2023), landscape-scale fuel-treatment planning (Altamimi et al. 2022), and grid-based suppression agents on Cell2Fire (Shen and Curtis 2022). Most use simplified fire dynamics without validated physics, single-agent formulations, or toy grids that lack realistic infrastructure; the closest prior system, Firehose (Shen and Curtis 2022), is single-objective (burned-area) and single-agent. To our knowledge, no prior work targets *multi-agent, infrastructure-weighted* suppression on a validated physics engine, which is the gap our benchmark addresses and on which we evaluate our proposed hierarchical method. Operations-research approaches remain the deployed state of the art for suppression *planning*: they optimize firebreak or resource placement via mixed-integer programming (Hu and Ntaimo 2009) under global observability and quasi-static fire, and are complementary to—not competitors of—the sequential, partially observed, reactive-suppression setting we study. Disaster-response RL more broadly has explored resource allocation (Nguyen, Nguyen, and Nahavandi 2020) but rarely the spatial coordination of multi-agent infrastructure defense under physically grounded fire spread.

**Cooperative MARL and Communication.** Centralized Training with Decentralized Execution (CTDE) is the dominant paradigm for cooperative MARL. QMIX (Rashid et al. 2018) factorizes joint value functions under monotonicity constraints; MAPPO (Yu et al. 2022) applies PPO (Schulman et al. 2017) with centralized critics. CommNet (Sukhbaatar, Szlam, and Fergus 2016) and TarMAC (Das et al. 2019) add learned communication channels. On our benchmark the effectiveness of these decentralized and communication-based methods varies by region and neither dominates; our value-aware hierarchical method, which pairs asset-value-driven assignment with a learned tactical policy, outperforms both. An ablation (Section 7) further finds that explicit communication does not, on its own, provide a statistically significant benefit at the three-agent scale we study.

**Hierarchical Reinforcement Learning.** The options framework (Sutton, Precup, and Singh 1999) and feudal networks (Vezhnevets et al. 2017) introduce temporal abstraction via subgoal hierarchies. HAM (Parr and Russell 1997) and MAXQ (Dietterich 1998) decompose MDPs into subtask hierarchies. In multi-agent settings, RODE (Wang et al. 2021) uses role decomposition, while MAVEN (Mahajan et al. 2019) conditions on shared latent variables. Most prior HRL work jointly optimizes a learned low-level policy with the high-level controller. Our approach pairs a *value-aware* strategic assignment (each agent to a high-value threatened asset) with a *learned* communicating tactical policy that is behavior-cloned from a value-aware asset-defense expert and

then refined by RL; we attribute performance to each component through ablation.

**Imitation Learning and RL Fine-Tuning.** Combining imitation with RL is well-established: DAgger (Ross, Gordon, and Bagnell 2011) iteratively corrects distributional shift; behavioral cloning followed by RL fine-tuning has been applied to game playing (Silver et al. 2016) and robotics (Rajeswaran et al. 2018). KL-regularized policy optimization prevents catastrophic forgetting of imitation priors (Wu, Tucker, and Nachum 2019). We apply this paradigm to the *tactical* layer of a hierarchical wildfire-coordination policy, using a value-aware asset-defense heuristic as the behavior-cloning teacher.

### 3 Problem Formulation

We model wildfire suppression as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) extended with hierarchical temporal abstraction.

**State.** The global state  $s = \langle S_{\text{fire}}, S_{\text{asset}}, S_{\text{agent}} \rangle$  consists of:

- $S_{\text{fire}} \in \{0, 1\}^{H \times W}$ : binary active-fire grid evolved by Cell2Fire physics.
- $S_{\text{asset}} \in \mathbb{R}_{\geq 0}^{H \times W}$ : infrastructure criticality map encoding economic asset values.
- $S_{\text{agent}} = \{(y_i, x_i)\}_{i=1}^N$ : grid positions of  $N$  suppression agents.

**Observations.** Each agent  $i$  observes an egocentric  $C \times C$  crop ( $C = 9$ ) containing 8 channels: active fire, treated cells, fuel type, criticality, agent occupancy, normalized coordinates, fire fraction, and time fraction. Agents do *not* observe the full grid.

**Actions.** The problem is hierarchical in time. At the *strategic* level, every  $T_{\text{macro}} = 10$  steps each agent is assigned a target—a high-value threatened asset to defend. At the *tactical* level, micro-actions  $a_i^t \in \{\text{stay, up, down, left, right, treat}\}$  are executed at every environment step to reach and defend that target.

**Reward.** The suppression objective, used both to train and to evaluate, rewards protecting economic value:

$$R = -\Delta \text{WEL} + \alpha \cdot \Delta \text{ISR} \quad (1)$$

where  $\text{WEL} = \sum_j v_j \cdot \mathbf{1}[\text{asset}_j \text{ burned}]$  is the Weighted Economic Loss,  $\text{ISR} = |\{\text{surviving assets}\}|/|\{\text{total assets}\}|$  is the Infrastructure Survival Rate, and  $\alpha$  upweights survival (Appendix A). This objective directly incentivizes infrastructure defense over generic burned-area minimization.

### 4 Methodology

**HierComm** decomposes infrastructure defense into a value-aware strategic layer and a learned tactical layer (Figure 1). Every  $T_{\text{macro}} = 10$  steps, a value-aware commander assigns each agent a high-value threatened asset to defend; between assignments, a learned communicating tactical policy selects micro-actions that build the firebreak protecting the assigned asset. The tactical policy is warm-started by behavior cloning and then refined end-to-end by reinforcement learning on the suppression objective (Eq. 1).

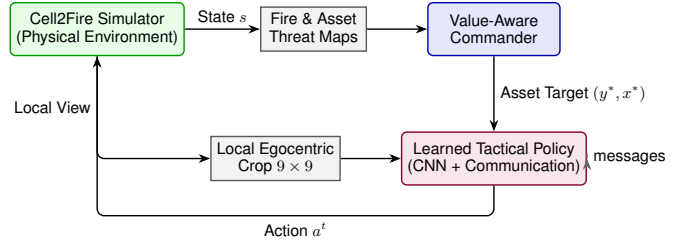


Figure 1: Architecture of the Value-Aware Hierarchical Reinforcement Learning framework.

#### 4.1 Value-Aware Strategic Commander

The commander assigns each agent a target asset by threat-weighted value. For every asset  $j$  with economic value  $v_j$ , it computes a threat score  $\tau_j = v_j / (1 + d_j)$ , where  $d_j$  is the distance from asset  $j$  to the nearest active-fire cell; agents are assigned to the highest-threat assets (ranked by  $\tau_j$ , or by value when no fire is present). The assigned asset location becomes the agent’s strategic target  $(y_i^*, x_i^*)$ , re-evaluated every  $T_{\text{macro}}$  steps. Prioritizing valuable assets that are *imminently threatened*—rather than merely nearby fire—is the strategic component of HierComm; we isolate its contribution by ablation (Section 7). We also implement a learned-commander variant (a network over coarse fire/asset/agent grids, warm-started and updated by policy gradient jointly with the tactical layer); **all HierComm results we report use the value-aware assignment above**, and the learned-commander variant is evaluated only in the ablation.

#### 4.2 Learned Communicating Tactical Policy

Given its assigned target, each agent selects micro-actions with a shared-parameter neural policy. Agent  $i$  (i) encodes its egocentric  $9 \times 9$ , 8-channel observation with a small CNN into a feature  $h_i \in \mathbb{R}^{64}$ ; (ii) encodes the relative direction to its target  $(\Delta y_i, \Delta x_i) \in [-1, 1]^2$  and fuses it with  $h_i$ ; (iii) exchanges  $R = 2$  rounds of mean-pooled messages with teammates,

$$h_i \leftarrow \tanh(W_h h_i + W_c \bar{m}_{-i}), \quad \bar{m}_{-i} = \frac{1}{N-1} \sum_{k \neq i} h_k, \quad (2)$$

and (iv) maps the result to masked logits over  $\{\text{stay, up, down, left, right, treat}\}$ . Parameters are shared across agents, so the policy is agent-count agnostic. The message-passing rounds in Eq. 2 are the explicit *communication* component of the architecture; we measure their contribution directly in the ablation rather than assuming it.

#### 4.3 Training: BC Warm-Start then PPO

The suppression reward is sparse—economic loss registers only when fire reaches an asset, many steps after the actions that determined the outcome—so we train the tactical policy in two phases.

**Phase 1: Behavior-cloning warm-start.** We clone a value-aware *asset-defense* expert: given an agent’s assigned asset, the expert suppresses the fire-frontier cell nearest that

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**Algorithm 1** HierComm Training

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- 1: **Phase 1 (BC)**: collect value-aware asset-defense demonstrations; train tactical policy  $\pi_\phi$  by cross-entropy
  - 2: Initialize centralized critic  $V_\psi$
  - 3: **for** iteration = 1 to  $T$  **do**
  - 4:   Collect rollouts: every  $T_{\text{macro}}$  steps, the value-aware commander assigns each agent a threat-weighted asset;  $\pi_\phi$  acts with  $R$  communication rounds
  - 5:    $r_t \leftarrow -\Delta\text{WEL} + \alpha \Delta\text{ISR} + \eta (\text{burning}_{t-1} - \text{burning}_t)$
  - 6:   Compute GAE advantages using  $V_\psi$
  - 7:   Update  $\pi_\phi, V_\psi$  by clipped PPO with an entropy bonus
  - 8: **end for**
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asset—building the firebreak that protects it—and otherwise advances toward it. Cloning this value-aware defense, rather than value-blind nearest-fire suppression, initializes the tactical policy with asset-directed behavior. The tactical policy is trained by cross-entropy on the expert’s micro-actions over 30 demonstration episodes.

**Phase 2: PPO fine-tuning.** The warm-started policy is refined with PPO (Schulman et al. 2017) on the suppression objective (Eq. 1), using a centralized critic and generalized advantage estimation. A light firebreak-shaping term (coefficient  $\eta$ ), proportional to the per-step reduction in actively burning cells, densifies the otherwise sparse signal. Training runs for 100k environment steps per seed; the value-aware commander is fixed during tactical learning, so reinforcement learning refines tactical execution rather than re-deriving the assignment. Hyperparameters (communication rounds  $R=2$ , PPO clip 0.2, GAE  $\lambda=0.95$ , entropy and shaping coefficients) match the released checkpoints (Appendix A).

## 5 Experimental Setup

**Wildfire Simulator.** We use Cell2Fire (Pais et al. 2021), a validated wildfire spread simulator implementing the Canadian FBP system. Fire evolves on  $32 \times 32$  grids with heterogeneous fuel types, wind direction, and topographic slope effects.

**Infrastructure Topographies.** We evaluate on two real-world GIS-derived topographies:

- **Saudi Arabia (Eastern Province)**: Petroleum refineries and pipeline junctions with high localized economic values.
- **California (WUI)**: Scattered residential structures typical of Wildland-Urban Interface zones.

**Agents and Baselines.**  $N = 3$  suppression agents operate simultaneously. We compare:

- **No-Op**: agents take no suppression actions.
- **Value-First**: hand-crafted heuristic that assigns agents to high-value threatened assets; its assignment rule also drives HierComm’s value-aware commander.
- **Greedy-Risk**: assigns agents to the highest fire-load sectors.
- **Local Reactive**: routes each agent to the nearest treatable fire-frontier cell.

- **Flat MARL (MAPPO)**: decentralized PPO agents without strategic coordination.
- **CommNet**: decentralized agents with learned inter-agent communication.
- **HierComm (ours)**: our value-aware hierarchical method with a learned tactical policy (Section 4, Algorithm 1).

### Metrics.

- **Weighted Economic Loss (WEL  $\downarrow$ )**: Total value of burned infrastructure.
- **Infrastructure Survival Rate (ISR  $\uparrow$ )**: Fraction of assets unburned.
- **Coordination Efficiency (CE  $\uparrow$ )**: Fraction of unique cells treated (higher = less spatial overlap).
- **Total Reward ( $\uparrow$ )**: Cumulative shaped reward.

**Statistical Protocol.** Learned methods use **5 independent training seeds**; each checkpoint is evaluated on a held-out ignition stream (**15 episodes per seed group**), and the *per-training-seed mean* is the unit of analysis ( $n = 5$ ). Heuristic policies are deterministic given the evaluation stream. We report means with bootstrap 95% confidence intervals and Welch’s  $t$ -tests with effect sizes (Cohen’s  $d$ , pooled SD; omitted where a distribution is degenerate). The learned methods—MAPPO, CommNet, and HierComm—are trained for 100k environment steps on the same WEL/ISR objective used for evaluation (Appendix A, with training curves in Figure 3), removing any objective-level confound.

## 6 Results

Table 1 reports the multi-seed results. We report WEL and ISR, the infrastructure-relevant metrics; the shaped training reward is dominated by a policy-independent burned-area term and is not comparable across policies. Three findings emerge.

**Finding 1: HierComm achieves the best infrastructure protection in both regions.** HierComm attains the lowest WEL and highest ISR in both Saudi (WEL 7.02, ISR 0.809) and California (WEL 5.88, ISR 0.832). It improves significantly in WEL on every heuristic and on the learned communication baseline: versus CommNet,  $p = 0.048$  (Saudi) and  $p = 0.016$  (California); versus reactive coverage,  $p = 0.010$  (Saudi) and  $p < 0.001$  (California); and versus the Value-First heuristic,  $p < 0.001$  in both regions. In Saudi its margin over the strongest flat baseline (MAPPO, 9.77) is *not* statistically significant ( $p = 0.32$ ), though HierComm is better on the seed means; in California it significantly outperforms MAPPO ( $p = 0.002$ ). We therefore claim a decisive win on California and a best-on-average, statistically-tied-with-MAPPO result on Saudi. On the seed means, no baseline matches HierComm’s combination of low loss and high survival across both regions. Figure 2 illustrates the difference on a representative episode: HierComm contains the fire and defends its assets, while the baselines and inaction let it overrun the landscape.

Table 1: **Main results** on the suppression-relevant regime (5 training seeds; mean  $\pm$  std over seeds, heuristics deterministic). Lower WEL / higher ISR is better; **bold** marks the best per region. HierComm attains the best WEL and ISR in both regions; the strongest *baseline* differs by region—flat MARL on concentrated Saudi assets, reactive coverage on scattered California assets.

| Region              | Policy                 | WEL $\downarrow$                  | ISR $\uparrow$                      |
|---------------------|------------------------|-----------------------------------|-------------------------------------|
| <b>Saudi Arabia</b> | No-Op                  | 27.00 $\pm$ 0.00                  | 0.286 $\pm$ 0.000                   |
|                     | Value-First (teacher)  | 18.84 $\pm$ 0.00                  | 0.545 $\pm$ 0.000                   |
|                     | Greedy-Risk            | 24.72 $\pm$ 0.00                  | 0.358 $\pm$ 0.000                   |
|                     | Local Reactive         | 11.91 $\pm$ 0.00                  | 0.720 $\pm$ 0.000                   |
|                     | Flat MARL (MAPPO)      | 9.77 $\pm$ 5.10                   | 0.692 $\pm$ 0.158                   |
|                     | CommNet                | 12.44 $\pm$ 4.32                  | 0.682 $\pm$ 0.106                   |
|                     | <b>HierComm (ours)</b> | <b>7.02 <math>\pm</math> 2.36</b> | <b>0.809 <math>\pm</math> 0.058</b> |
| <b>California</b>   | No-Op                  | 34.00 $\pm$ 0.00                  | 0.000 $\pm$ 0.000                   |
|                     | Value-First (teacher)  | 28.40 $\pm$ 0.00                  | 0.156 $\pm$ 0.000                   |
|                     | Greedy-Risk            | 28.40 $\pm$ 0.00                  | 0.156 $\pm$ 0.000                   |
|                     | Local Reactive         | 10.23 $\pm$ 0.00                  | 0.702 $\pm$ 0.000                   |
|                     | Flat MARL (MAPPO)      | 14.17 $\pm$ 2.89                  | 0.624 $\pm$ 0.110                   |
|                     | CommNet                | 19.58 $\pm$ 7.77                  | 0.398 $\pm$ 0.248                   |
|                     | <b>HierComm (ours)</b> | <b>5.88 <math>\pm</math> 1.04</b> | <b>0.832 <math>\pm</math> 0.027</b> |

**Finding 2: The learned policy substantially outperforms its heuristic teacher.** HierComm is warm-started from the Value-First heuristic and then refined by reinforcement learning. Rather than merely reproducing its teacher, the refined policy *surpasses* it by a wide margin: WEL 7.02 vs. 18.84 on Saudi and 5.88 vs. 28.40 on California (both  $p < 0.001$ , large effect sizes). The learned tactical layer and RL fine-tuning—not the heuristic dispatch alone—are responsible for this gain, which our component ablation isolates.

**Finding 3: The benchmark remains diagnostic; HierComm generalizes across it.** No single baseline is strongest in both regions: flat MARL (MAPPO) is the best baseline on concentrated Saudi assets (WEL 9.77), whereas reactive coverage is best on scattered California assets (10.23); the learned communication baseline (CommNet) is competitive on Saudi (12.44) yet among the weakest learned methods on California (19.58). Concentrated petroleum assets and scattered WUI assets reward different coordination strategies, and a single-region evaluation would conflate them—the benchmark’s central diagnostic property. HierComm is the only method best (on average) in *both* regions, indicating that value-aware hierarchical control performs consistently across the two studied asset geometries rather than exploiting one; whether this consistency extends to further geometries is an open question (Section 10).

## 7 Ablation Analysis

To identify which components of HierComm are responsible for its performance, we remove one component at a time and re-evaluate under the main-results protocol (Table 2). Each ablated model is otherwise identical to the deployed HierComm, whose scores appear in the *Full* row. We test each variant against the full model with Welch’s  $t$ -test on the per-seed WEL. Replacing the tactical policy with deterministic target-seeking (“w/o learned tactical”) removes all learned

Table 2: **Component ablation of HierComm** (5 training seeds; seed-mean WEL/ISR), ordered by the effect of removing each component. Removing the learned tactical layer, the hierarchy, or RL fine-tuning significantly degrades performance; removing communication or shaping does not (Welch  $p$  vs. full on WEL reported in text).

| Variant                    | Saudi            |                | California       |                |
|----------------------------|------------------|----------------|------------------|----------------|
|                            | WEL $\downarrow$ | ISR $\uparrow$ | WEL $\downarrow$ | ISR $\uparrow$ |
| Full HierComm              | 7.02             | 0.809          | 5.88             | 0.832          |
| w/o learned tactical       | 18.76            | 0.465          | 13.76            | 0.451          |
| w/o hierarchy (flat comms) | 12.44            | 0.682          | 19.58            | 0.398          |
| w/o RL fine-tune           | 11.33            | 0.696          | 9.46             | 0.721          |
| w/o communication          | 5.67             | 0.860          | 4.66             | 0.864          |
| w/o tactical shaping       | 5.41             | 0.840          | 7.60             | 0.787          |

parameters and yields a single deterministic score, so no seed-variance test is reported for that row.

**Three components account for the gains.** The **learned tactical policy** is the largest contributor: replacing it with deterministic target-seeking under the same value-aware commander raises WEL from 7.02 to 18.76 (Saudi) and 5.88 to 13.76 (California), roughly halving ISR. Removing the **value-aware hierarchy**—leaving a flat communicating policy (CommNet)—raises WEL to 12.44 (Saudi,  $p = 0.048$ ) and 19.58 (California,  $p = 0.016$ ). Removing **RL fine-tuning**—using the behavior-cloned policy without PPO—raises WEL to 11.33 (Saudi,  $p = 0.011$ ) and 9.46 (California,  $p < 0.001$ ). Each removal significantly worsens infrastructure protection in both regions, and each lowers ISR correspondingly.

**Communication: an informative negative result.** Removing **explicit communication** (setting the message-

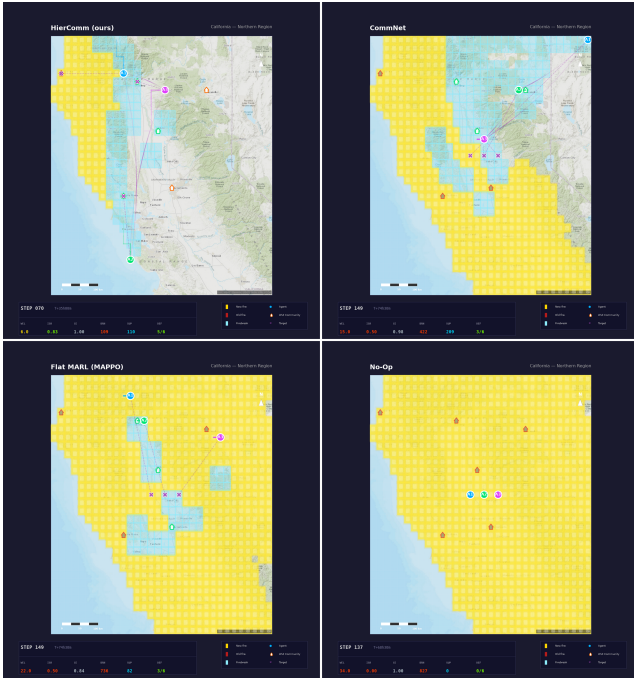


Figure 2: **Qualitative comparison** on a single representative California episode (seed 42), shown at episode end. HierComm (top-left) concentrates suppression to contain the fire and keeps most infrastructure (house markers) unburned, whereas CommNet, Flat MARL (MAPPO), and No-Op let the fire spread across the landscape. On-frame WEL/ISR are for this single episode; aggregate results over all seeds and episodes are in Table 1.

passing rounds to zero) produces no statistically significant change in WEL (Saudi  $p = 0.30$ ; California  $p = 0.12$ ); the small numerical differences fall within seed variance, so we treat communication as having *no measurable effect* in either direction rather than helping or hurting. Removing the **firebreak-shaping** term is likewise not significant ( $p = 0.24$  Saudi,  $p = 0.06$  California). We report this plainly: in the evaluated three-agent benchmark, the value-aware hierarchy, learned tactical control, and RL refinement account for the observed gains, while explicit message passing provides no statistically significant additional benefit. One possible explanation—which our experiments do not directly test—is structural: with a value-aware commander already assigning complementary asset targets to only three agents, some of the coordination that message passing might supply may already be provided by the hierarchy. Communication remains part of the architecture; we simply do not claim it as a source of improvement.

The components that account for these gains are also what sustain HierComm’s robustness (Section 8): because the value-aware commander re-derives asset targets from the current fire and asset maps, the learned tactical policy keeps defending the right assets when conditions shift.

Table 3: **Difficulty-regime robustness** (seed-mean WEL ↓). HierComm is best in every regime; No-Op worsens monotonically with difficulty.

| Region     | Policy      | Easy       | Medium     | Hard        |
|------------|-------------|------------|------------|-------------|
| Saudi      | No-Op       | 26.8       | 27.0       | 27.0        |
|            | Value-First | 18.0       | 18.8       | 19.4        |
|            | CommNet     | 4.2        | 12.4       | 18.0        |
|            | HierComm    | <b>3.3</b> | <b>7.0</b> | <b>11.3</b> |
| California | No-Op       | 12.7       | 34.0       | 34.0        |
|            | Value-First | 7.0        | 28.4       | 32.8        |
|            | CommNet     | 8.1        | 19.6       | 25.6        |
|            | HierComm    | <b>0.4</b> | <b>5.9</b> | <b>17.7</b> |

## 8 Robustness and Transfer

Strong average performance is only useful if it holds under conditions the policy was not trained on. We evaluate HierComm along three axes: difficulty regimes, held-out perturbations, and cross-region transfer. Held-out and transfer results use a falsifiable metric,  $\Delta\text{WEL} = \text{WEL}(\text{No-Op}) - \text{WEL}(\text{policy})$  computed *within* each condition, so a policy that adds no value scores 0 and cannot hide behind a favorable baseline.

**Difficulty regimes.** We re-evaluate the frozen policies under easy/medium/hard regimes that vary fire weather and ignition proximity (Table 3). No-Op WEL rises monotonically with difficulty in both regions, confirming the axis is meaningful, and HierComm attains the lowest WEL in every regime, degrading gracefully (Saudi  $3.3 \rightarrow 7.0 \rightarrow 11.3$ ; California  $0.4 \rightarrow 5.9 \rightarrow 17.7$ ) while remaining ahead of every baseline. The ranking is thus stable across difficulty rather than an artifact of one operating point.

**Held-out conditions.** We perturb the trained environment along three axes it never saw during training—ignitions forced to unlikely map-corner cells, wind rotated  $90^\circ$ , and a  $90^\circ$ -rotated infrastructure layout—and measure  $\Delta\text{WEL}$  (Table 4). HierComm retains a clear protective margin in every case: it stays well above the No-Op line under rotated wind (+13.2 Saudi, +21.9 California) and, notably, under a rotated asset layout (+26.2 Saudi, +17.1 California), indicating that its value-aware defense follows the assets rather than memorizing their positions. The smallest margins are under held-out corner ignitions (+13.4/+13.1), which remain clearly positive.

**Cross-region transfer.** Transfer between the two regions is the hardest test, as the asset geometries differ sharply. When a policy trained on the other region is deployed here, HierComm still improves over No-Op ( $\Delta\text{WEL} +10.2$  evaluated on Saudi, +14.4 on California), but with clear degradation: its Infrastructure Survival Rate falls from native 0.81/0.83 to 0.55 (California→Saudi) and 0.38 (Saudi→California)—a Transfer-Robustness-Score (transferred/native ISR) of 0.66 and 0.47. The comparison with CommNet is instructive rather than uniformly favorable: CommNet transfers slightly

Table 4: **Held-out and transfer robustness** ( $\Delta\text{WEL} = \text{WEL}(\text{No-Op}) - \text{WEL}(\text{HierComm})$  within each condition; higher is better, 0 means no better than inaction).

| Condition         | Saudi | California |
|-------------------|-------|------------|
| Standard          | 20.0  | 28.1       |
| Held-out ignition | 13.4  | 13.1       |
| Wind +90°         | 13.2  | 21.9       |
| Rotated assets    | 26.2  | 17.1       |
| Cross-region      | 10.2  | 14.4       |

better in one direction (California→Saudi, TRS 0.96) but collapses in the other (Saudi→California, TRS 0.10), whereas HierComm avoids catastrophic failure in either direction. HierComm thus offers more *consistent* transfer, not uniformly higher transfer; cross-region generalization remains a limitation for all methods and a target for future work.

**Practical implication.** Across difficulty and held-out perturbations—and, to a lesser degree, cross-region deployment—HierComm never drops to the inaction baseline, a consistency that matters for infrastructure protection alongside peak performance.

## 9 Discussion

HierComm composes established ingredients—a hand-designed value-aware assignment rule, a CommNet-style tactical actor, behavior cloning, and PPO—and we claim no algorithmic novelty in these parts; its value is that this simple composition is strong and reproducible, and that the ablation localizes *why* it works (the value-aware hierarchy and the RL-refined tactical policy, not message passing). We thus regard the benchmark and this analysis as the primary contributions, with the method a transparent reference point. The outcome is region-dependent by design: on concentrated Saudi assets several baselines already reach relatively low loss, so the achievable separation is small and HierComm’s margin over the strongest flat baseline is within seed variance, whereas the decisive, statistically significant separation appears on scattered California assets, where value-blind strategies fail to prioritize among dispersed targets. Because every learned method optimizes the identical WEL/ISR objective and the regime was calibrated independently of HierComm (Section 1), the comparison isolates coordination quality rather than a tuned objective or environment; significance nonetheless rests on  $n = 5$  seed means, which we read conservatively.

## 10 Limitations

We acknowledge several limitations:

**Communication shows no measurable benefit at this scale.** Our ablation finds that explicit inter-agent message passing does not significantly affect performance in the three-agent setting studied (Section 7). Whether communication becomes beneficial at larger team sizes, with denser interactions, or under a different message design is an open question we do not resolve here.

**Fixed strategic assignment.** The deployed HierComm uses a value-aware assignment rule rather than a learned commander (the learned-commander variant did not improve results). A rule-based strategic layer is transparent but may limit adaptability in situations its heuristic does not anticipate; learning the strategic layer productively remains future work.

**Reward–metric coupling and evaluation scope.** The tactical policy is trained on the same WEL/ISR quantities used for evaluation; while all learned baselines share this objective (removing relative confound), absolute performance should be read with that coupling in mind. All results use a single simulator (Cell2Fire), a single month of weather, and asset-criticality rasters that are stylized rather than economically measured (Appendix G); cross-simulator and multi-season validation remain open.

**Two landscapes and small-sample statistics.** Our conclusions rest on two hand-constructed regions and  $n = 5$  training seeds. Two geometries cannot establish generalization to arbitrary asset layouts—we accordingly scope claims to the studied geometries—and several tests rest on five seed means. We therefore report effect sizes alongside  $p$ -values and read marginal results (e.g., HierComm vs. the communicating baseline on Saudi,  $p = 0.048$ ) cautiously; the single largest ablation effect, removing the learned tactical layer, is a deterministic policy with no seed variance and is reported by magnitude rather than a test. Broader validation across more regions, seeds, and simulators is future work.

**Scale and transfer.** We evaluate primarily at  $N = 3$  agents on  $32 \times 32$  grids. The tactical policy shares parameters across agents and can run at other team sizes, but we have not tuned or validated larger teams; larger grids ( $64^2+$ ) require rebuilding the infrastructure rasters and retraining. Cross-region transfer degrades markedly relative to native performance for all methods (transfer-robustness 0.47–0.66 for HierComm; Section 8); HierComm avoids the catastrophic one-directional collapse of the communicating baseline but is not a strong zero-shot transfer method, so a policy trained on one asset geometry should not be deployed on a very different one without adaptation.

**Fire dynamics.** Each agent step advances the simulator half a simulated hour and the fire evolves every step; nonetheless the episode horizon ( $\sim 75$  hours) captures a single fire’s active phase rather than a multi-day campaign.

## 11 Broader Impacts

Wildfire suppression research has clear positive potential: effective coordination can save lives and protect critical infrastructure. However, we emphasize that our system operates on a *simulated* environment and is not validated for real-world deployment. Autonomous wildfire suppression decisions could lead to negative outcomes if deployed without human oversight, particularly in scenarios involving civilian evacuation or resource triage. We advocate for AI-assisted decision support rather than fully autonomous suppression.

## 12 Conclusion

We presented an infrastructure-aware wildfire-coordination benchmark on validated Cell2Fire physics with two documented GIS landscapes, together with **HierComm**, a value-aware hierarchical system that pairs a rule-based asset-value assignment with a tactical policy learned by imitation and RL fine-tuning. Under a uniform, reproducible 5-seed protocol, HierComm attains the best average WEL and ISR in both regions—a decisive, statistically significant margin on the scattered-asset (California) landscape and a best-on-average result that is statistically tied with flat MAPPO on the concentrated-asset (Saudi) landscape—and remains protective across difficulty regimes, held-out perturbations, and, with degradation, cross-region transfer. We claim no novelty for the underlying primitives; our contributions are the benchmark, the integrated system, and an honest component analysis showing that the value-aware hierarchy and RL-refined tactical policy—not explicit communication at this scale—drive performance. We release the code, checkpoints, GIS pipeline, and hashed manifests to support future work on learned coordination for spatially critical infrastructure defense.

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## Appendix

### A. Hyperparameters

Table 5: Training and architecture hyperparameters.

| Parameter                                               | Value                                    |
|---------------------------------------------------------|------------------------------------------|
| <b>Environment</b>                                      |                                          |
| Grid size                                               | $32 \times 32$                           |
| Number of agents                                        | 3                                        |
| Crop size (egocentric observation)                      | $9 \times 9$                             |
| Max episode steps                                       | 150                                      |
| Simulated minutes per agent step                        | 30 (fire advances every step)            |
| Coordination penalty                                    | 0.1                                      |
| Infrastructure catastrophe weight                       | 2.0                                      |
| Wrapper cascade probability                             | 0.1                                      |
| Reward ISR weight ( $\alpha$ , Eq. 1)                   | 20                                       |
| <b>HierComm: strategic commander (value-aware rule)</b> |                                          |
| Macro-step interval ( $T_{\text{macro}}$ )              | 10                                       |
| Assignment                                              | threat-weighted asset, $v_j / (1 + d_j)$ |
| <b>HierComm: tactical policy</b>                        |                                          |
| CNN feature dim                                         | 64                                       |
| Target-encoding dim                                     | 16                                       |
| Communication rounds ( $R$ )                            | 2                                        |
| <b>HierComm: BC warm-start (tactical)</b>               |                                          |
| BC demonstration episodes                               | 30                                       |
| BC epochs / learning rate                               | $60 / 1 \times 10^{-3}$                  |
| <b>HierComm: PPO fine-tuning (tactical)</b>             |                                          |
| Training environment steps                              | 100,000                                  |
| Actor / critic learning rate                            | $3 \times 10^{-4} / 1 \times 10^{-3}$    |
| PPO clip / GAE $\lambda$                                | 0.2 / 0.95                               |
| PPO epochs / batch size                                 | 4 / 256                                  |
| Entropy coef / shaping coef                             | 0.02 / 0.05                              |
| Optimizer                                               | Adam                                     |
| <b>Decentralized baselines (MAPPO / CommNet)</b>        |                                          |
| Training environment steps                              | 100,000                                  |
| Training objective                                      | WEL/ISR (Eq. 1)                          |
| Actor CNN features / critic hidden                      | 64 / 128                                 |
| MAPPO GAE $\lambda$ / PPO clip                          | 0.95 / 0.2                               |
| CommNet communication rounds                            | 2                                        |

Values are extracted from the released checkpoint configurations. The deployed HierComm uses the value-aware assignment rule (no learned commander parameters); the learned-commander variant evaluated only in the ablation adds a  $4 \times 4$ -grid commander MLP (hidden dim 128) updated by policy gradient (learning rate  $5 \times 10^{-5}$ , KL coefficient 0.3). Models are trained over five seeds {42, 1042, 2042, 3042, 4042} and evaluated on a held-out ignition stream (15 episodes per seed group). Compute and software versions are disclosed in Appendix F.

### B. Per-Seed Results

### C. Statistical Test Details

Comparisons use the per-seed means ( $n = 5$ ). Against the deterministic heuristics (e.g. Value-First) we use a one-sample  $t$ -test; against learned baselines, Welch’s  $t$ -test. Cohen’s  $d$  is pooled-SD (sign: policy – reference on WEL, so negative  $d$  = the policy has lower WEL, i.e. is better).

HierComm significantly improves in WEL over the Value-First teacher and CommNet in both regions, and over Flat MARL (MAPPO) in California; its advantage over MAPPO on Saudi is on the seed means but not statistically significant.

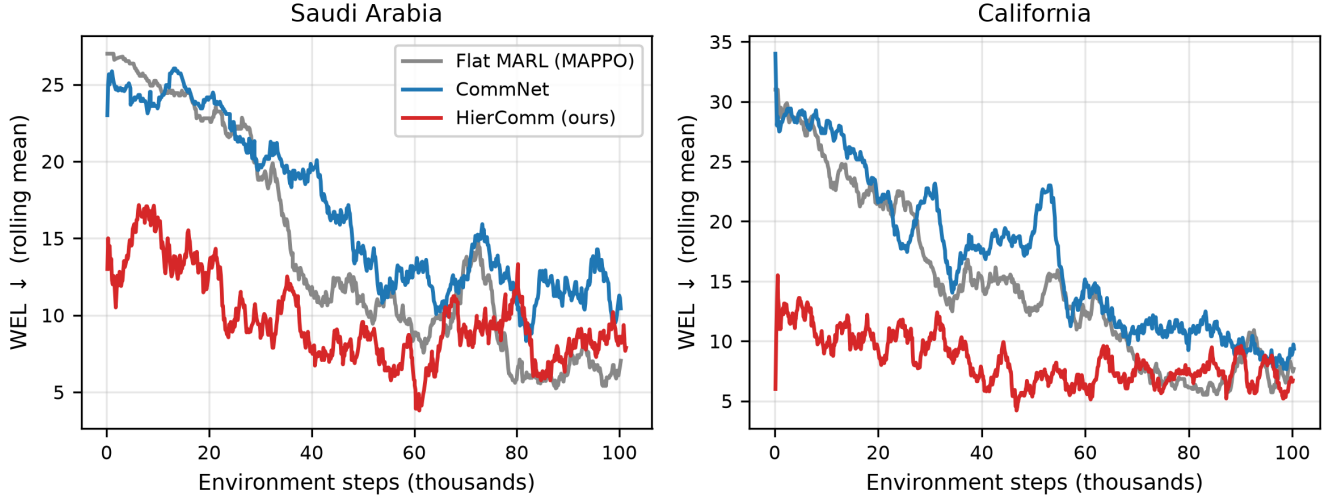


Figure 3: **Learned-method training curves** (representative seed 42; WEL rolling mean over 25 episodes, 100k environment steps on the WEL/ISR objective). WEL decreases over training for the decentralized baselines and HierComm and converges within the budget—evidence that the learned methods are trained to convergence rather than undertrained. HierComm starts low owing to its behavior-cloning warm-start.

Table 6: Per-seed WEL over the five training seeds {42, 1042, 2042, 3042, 4042} for HierComm and the CommNet baseline; the per-seed mean is the unit of analysis.

| Policy, Region       | S1   | S2    | S3    | S4    | S5    |
|----------------------|------|-------|-------|-------|-------|
| HierComm, Saudi      | 6.44 | 10.44 | 4.61  | 5.33  | 8.29  |
| CommNet, Saudi       | 9.11 | 13.67 | 11.84 | 8.41  | 19.17 |
| HierComm, California | 6.16 | 4.89  | 6.48  | 7.13  | 4.72  |
| CommNet, California  | 7.84 | 21.00 | 17.36 | 22.79 | 28.89 |

## D. Checkpoint Hashes

## E. Simulator: Cell2Fire Physics vs. Wrapper

To avoid overstating what is physically validated, we delineate exactly which components come from Cell2Fire and which are our custom wrapper.

**Cell2Fire (validated physics).** The fire spread itself is computed by the Cell2Fire C++ engine (Pais et al. 2021), which implements the Canadian Forest Fire Behavior Prediction (FBP) System: FBP fuel types (mapped from remote-sensing NDVI; Appendix G), wind- and slope-driven directional rate of spread, one fire period = 1 simulated minute, and stochastic spread governed by the rate-of-spread coefficient of variation. We invoke it through an interactive stdin/stdout step protocol.

**Custom wrapper (not part of Cell2Fire/FBP).** The following are our additions and should not be read as validated fire physics: (i) the *suppression/treatment* action model; (ii) the infrastructure *cascade* model (per-neighbor ignition probability 0.1 when an asset cell burns) — the “cascade probability” of Appendix A is a wrapper parameter, not an FBP concept; (iii) the infrastructure-weighted reward (Eq. 1); (iv) the PettingZoo `ParallelEnv` multi-agent layer with action masking (agents cannot move off-grid or treat non-fuel cells). Each agent step advances the simulator by 30 fire periods (half a simulated hour), so the fire evolves at every step.

## F. Compute and Reproducibility Disclosure

All experiments in this paper ran on a single workstation, CPU-only (no GPU/CUDA was used):

- **Hardware:** AMD Ryzen 5 5600H (6 cores / 12 threads),  $\approx$  7 GB RAM.
- **Software:** Linux (WSL2, kernel 6.6); Python 3.12.3; PyTorch 2.12.1, NumPy 1.26.4, Gymnasium 1.0.0, PettingZoo 1.26.1, SciPy 1.17.1, pandas 2.3.3, Matplotlib 3.11.0.
- **Wall-clock:** each decentralized baseline trains for 100k environment steps in  $\approx$  1.7 h (California) to 2.5 h (Saudi); a full 5-seed  $\times$  15-episode evaluation of all policies takes  $\approx$  1.5 h per region.

Table 7: Key WEL comparisons (HierComm vs. baselines) under the uniform protocol.

| Region     | Comparison               | $t$    | $p$                  | $d$    |
|------------|--------------------------|--------|----------------------|--------|
| Saudi      | HierComm vs. Value-First | -11.19 | $3.6 \times 10^{-4}$ | -5.01  |
|            | HierComm vs. Flat MARL   | -1.09  | 0.319                | -0.69  |
|            | HierComm vs. CommNet     | -2.46  | 0.048                | -1.56  |
| California | HierComm vs. Value-First | -48.4  | $1.1 \times 10^{-6}$ | -21.65 |
|            | HierComm vs. Flat MARL   | -6.03  | $1.8 \times 10^{-3}$ | -3.81  |
|            | HierComm vs. CommNet     | -3.90  | 0.016                | -2.47  |

Table 8: SHA-256 hashes (first 16 characters) of the released seed-42 checkpoints for each learned policy. The full five-seed set (30 checkpoints) ships with a complete SHA-256 manifest. Non-learned policies (No-Op, Value-First, Greedy-Risk, Local Reactive) carry no checkpoint.

| Checkpoint                                 | SHA-256 (first 16 characters) |
|--------------------------------------------|-------------------------------|
| checkpoint_hiercomm_heur_saudi_s42.pt      | 9b839de90b48abff              |
| checkpoint_hiercomm_heur_california_s42.pt | 13c3dc0f0dd17de6              |
| checkpoint_mappo_saudi_s42.pt              | 20d2c9a757acf3f9              |
| checkpoint_mappo_california_s42.pt         | 5c371699b3208eb1              |
| checkpoint_commnet_saudi_s42.pt            | 009796e78da411fc              |
| checkpoint_commnet_california_s42.pt       | 31fd5ba2a0a3f922              |

- **Seeds:** training seed 42; evaluation seeds  $\{42, 1042, 2042, 3042, 4042\}$  with disjoint train/eval streams. All reported means are over the five evaluation-seed means ( $n = 5$ ).

A release manifest binds every checkpoint and result artifact to its SHA-256 hash.

## G. Data Provenance

The two regions are built from public remote-sensing sources over a June 2025 window; raw data is not redistributed but is downloaded from the providers and processed deterministically into  $32 \times 32$  Cell2Fire landscapes (converter `wildfire_marl.data.to_cell2fire`; every landscape carries a `conversion_report.json` binding it to its input hashes).

- **Regions / ROIs:** Saudi Eastern Province (lon/lat  $[45.5, 23.5, 50.5, 28.5]$ , desert regime) and Northern California (lon/lat  $[-124.5, 36.5, -119.0, 41.5]$ , forest/mountain). CRS EPSG:4326, north-up.
- **Fuels:** MODIS NDVI  $\rightarrow$  Canadian FBP fuel classes via fixed, region-independent thresholds (Forestry Canada Fire Danger Group 1992; Pais et al. 2021). A declared sensitivity parameter.
- **Weather:** ERA5 (Copernicus) June-2025 extracts  $\rightarrow$  hourly `Weather.csv`; FWI codes via the standard CFFDRS equations. Wind direction is the meteorological from-direction.
- **Topography:** SRTM DEM (public domain)  $\rightarrow$  per-cell slope percent and uphill azimuth in FBP convention.
- **Ignitions:** NASA FIRMS detections restricted to burnable cells, detection-count-weighted with disjoint train/eval seed streams.
- **Infrastructure criticality.** The asset-criticality raster is a *stylized* layer (Gaussian value falloff around public infrastructure locations), not measured economic valuations; it is a declared modeling choice, and asset values are a sensitivity parameter rather than ground truth.
- **Resolution abstraction.** `Forest.asc` declares Cell2Fire’s validated 100 m operating cell size while spatial patterns derive from coarser ROI tiles; absolute geographic distance is not preserved.
- **Known data debt.** The raw California NDVI GeoTIFF was not archived, so California NDVI is reconstructed under a declared linear assumption; this is covered by the fuel-map sensitivity analysis and flagged for closure by re-download.
- **Release.** Code, configs, checkpoints (Appendix D), and the generated landscapes are released with a SHA-256 manifest; a fixed snapshot may be archived on Zenodo with a DOI.